

# COMP0078: Supervised Learning

## Lecture 8 — Online Learning II

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*Revision notation: items marked with  $\star$  are specifically listed in the revision notes file (`revision25-26-V1.pdf`). All content in this file is examinable; Part II (not examined) was not included.*

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# 1 Online Learning with Partial Feedback (Bandits)\*

## 1.1 Full vs. Partial Feedback\*

**Definition 1.1** (Full Feedback Protocol). For  $t = 1$  to  $m$ :

1. Predict  $\hat{y}_t \in [n]$ .
2. Observe the *full* loss vector  $\ell_t \in [0, 1]^n$ .

**Goal.** Design master algorithms with sub-linear regret:

$$\sum_{t=1}^m \ell_{t, \hat{y}_t} - \min_{i \in [n]} \sum_{t=1}^m \ell_{t, i} \leq o(m). \quad (1)$$

**Definition 1.2** (Partial (Bandit) Feedback Protocol\*). For  $t = 1$  to  $m$ :

1. Predict  $\hat{y}_t \in [n]$ .
2. Observe *only* the loss of the chosen action  $\ell_{t, \hat{y}_t} \in [0, 1]$ .

The goal is the same as in Definition 1.1. Although only one component is observed, the entire loss vector  $\ell_t$  is assumed to exist.

**Example 1.3** (Bandit Applications).

1. **Online advertising:**  $n$  candidate ads; loss is 0 if the user clicks through.
2. **Medical trials:**  $n$  treatments; better patient response yields lower loss.
3. **Game-tree search:**  $n$  branches; loss decreases with evaluation-function improvement.

**Remark 1.4** (Exploration–Exploitation Trade-off\*). The bandit setting is named after slot machines (“one-armed bandits”):

1. *Exploration*: try different arms to estimate which has minimal expected loss.
2. *Exploitation*: play the arm with minimal observed loss.
3. The fundamental trade-off: trials used for exploration limit exploitation and vice versa.

## 1.2 Unbiased Estimators\*

**Definition 1.5** (Unbiased Estimator\*). An estimator  $\hat{\theta}$  estimates a parameter  $\theta$  of a distribution from a sample. It is **unbiased** if and only if  $\mathbb{E}[\hat{\theta}] = \theta$ .

**Example 1.6** (Sample Mean). Suppose  $X_1, \dots, X_n$  are IID random variables with mean  $\mu$ . Then  $\hat{\theta} := \frac{1}{n}(X_1 + \dots + X_n)$  is an unbiased estimator of  $\mu$  (by Definition 1.5).

**Example 1.7** (German Tank Problem). Suppose  $X$  is uniformly distributed over  $\{1, \dots, n\}$  with  $n$  unknown. The maximum-likelihood estimator  $\hat{\theta}_1 := X$  satisfies  $\mathbb{E}[\hat{\theta}_1; \theta = n] = \frac{n+1}{2}$ , so it is biased. The adjusted estimator  $\hat{\theta}_2 := 2X - 1$  is unbiased:

$$\mathbb{E}[\hat{\theta}_2; \theta = n] = \sum_{x=1}^n \frac{1}{n} (2x - 1) = 2 \cdot \frac{n+1}{2} - 1 = n.$$

## 1.3 Importance Weighting\*

**Remark 1.8** (Importance-Weighted Estimator\*). Let  $X$  be a real-valued random variable with mean  $\mu$ . Define a biased coin  $Z_p$  with  $\Pr(Z_p = H) = p$ . The estimator

$$\hat{\theta}_p := \begin{cases} X/p & \text{if } Z_p = H, \\ 0 & \text{if } Z_p = T, \end{cases}$$

satisfies  $\mathbb{E}[\hat{\theta}_p] = p \cdot (X/p) + (1 - p) \cdot 0 = X$  and is therefore unbiased (Definition 1.5).

**Definition 1.9** (Hallucinated Loss Vector\*). Given a distribution  $v_t \in \Delta_n$ , sample  $\hat{y}_t \sim v_t$ . The **hallucinated (importance-weighted) loss vector**  $\ell_t^h$  is

$$\ell_{t,i}^h := \frac{\ell_{t,i}}{v_{t,i}} \mathbf{1}[i = \hat{y}_t], \quad i \in [n]. \quad (2)$$

Only the entry  $i = \hat{y}_t$  is nonzero:  $\ell_t^h = (0, \dots, 0, \ell_{t,\hat{y}_t}/v_{t,\hat{y}_t}, 0, \dots, 0)$ . For every  $i \in [n]$ ,

$$\mathbb{E}_{\hat{y}_t \sim v_t}[\ell_{t,i}^h] = \sum_{j=1}^n v_{t,j} \frac{\ell_{t,i}}{v_{t,i}} \mathbf{1}[i = j] = \ell_{t,i},$$

so  $\ell_t^h$  is an unbiased estimator of  $\ell_t$  (Definition 1.5).

**Remark 1.10.** By observing a *single* arm one obtains a simultaneous unbiased estimate for *all* arms. Since we already have a full-information algorithm (HEDGE), the idea is to apply HEDGE to the hallucinated loss vectors. Two subtleties remain: (i)  $\ell_t^h$  can be unbounded, whereas HEDGE requires bounded losses; (ii) the regret bound will hold in expectation, requiring a careful adversarial model.

## 1.4 The EXP3 Algorithm\*

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**Algorithm 1** EXP3: Exponential-weight algorithm for Exploration and Exploitation [1]\*

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**Require:** Learning rate  $\eta \in (0, \infty)$

- 1:  $v_1 \leftarrow (1/n, \dots, 1/n)$
  - 2: **for**  $t = 1$  **to**  $m$  **do**
  - 3:   Sample  $\hat{y}_t \sim v_t$   $\triangleright$  treat  $v_t$  as a distribution over  $[n]$
  - 4:   Observe  $\ell_{t,\hat{y}_t} \in [0, 1]$
  - 5:   Construct hallucinated loss  $\ell_{t,i}^h := \frac{\ell_{t,i}}{v_{t,i}} \mathbf{1}[i = \hat{y}_t]$  for  $i \in [n]$   $\triangleright$  Definition 1.9
  - 6:    $v_{t+1,i} \leftarrow \frac{v_{t,i} \exp(-\eta \ell_{t,i}^h)}{Z_t}$  where  $Z_t = \sum_{j=1}^n v_{t,j} \exp(-\eta \ell_{t,j}^h)$
  - 7: **end for**
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## 1.5 Adversarial Model\*

**Definition 1.11** (Deterministic Oblivious Adversary\*). In this model the loss vectors  $\ell_1, \dots, \ell_m$  are determined *before* the algorithm begins. The adversary may have complete knowledge of the learner's algorithm and use this to set the losses, but it is *non-adaptive*: it cannot change the losses “on-the-fly” in response to the learner's random choices. This model subsumes the stochastic bandit model (sample  $D_1, \dots, D_m$  in advance).

## 1.6 EXP3 Analysis\*

**Lemma 1.12** (EXP3 Initial Inequality). *For any loss sequence  $\ell_1, \dots, \ell_m \in [0, 1]^n$ , Algorithm 1 (EXP3) satisfies*

$$\sum_{t=1}^m v_t \cdot \ell_t^h - \sum_{t=1}^m u \cdot \ell_t^h \leq \frac{\ln n}{\eta} + \frac{\eta}{2} \sum_{t=1}^m \sum_{i=1}^n v_{t,i} (\ell_{t,i}^h)^2 \quad \forall u \in \Delta_n. \quad (3)$$

*Proof.* The inequality follows because EXP3 is HEDGE applied with  $\ell_t$  replaced by  $\ell_t^h$  (Definition 1.9). The bound was established for HEDGE in Lecture 7 (“HEDGE Theorem – Proof (2)”).  $\square$

We now state the key intermediate identities used in the proof of the main theorem.

**Lemma 1.13** (Expected Inner Products). *Under the randomness  $\hat{y}_t \sim v_t$ :*

1.  $\mathbb{E}[\ell_{t,i}^h] = \ell_{t,i}$  for all  $i \in [n]$  (by Definition 1.9).

2.  $\mathbb{E}[v_t \cdot \ell_t^h] = \sum_{i=1}^n v_{t,i} \ell_{t,i} = \mathbb{E}[\ell_{t,\hat{y}_t}]$ .

3.  $\mathbb{E}[(\ell_{t,i}^h)^2] = \frac{\ell_{t,i}^2}{v_{t,i}}$ , whence

$$\mathbb{E}\left[\sum_{i=1}^n v_{t,i} (\ell_{t,i}^h)^2\right] = \sum_{i=1}^n \ell_{t,i}^2 \leq n. \quad (4)$$

**Theorem 1.14** (EXP3 Regret Bound [1]<sup>\*</sup>). *Denote  $L_A(S) := \sum_{t=1}^m \ell_{t,\hat{y}_t}$  and  $L_i(S) := \sum_{t=1}^m \ell_{t,i}$ . For any loss sequence  $S = \ell_1, \dots, \ell_m \in [0, 1]^n$  under the deterministic oblivious adversary (Definition 1.11), EXP3 (Algorithm 1) with  $\eta = \sqrt{2 \ln n / (mn)}$  achieves*

$$\mathbb{E}[L_A(S)] - \min_i L_i(S) \leq \sqrt{2mn \ln n}. \quad (5)$$

*Proof of Theorem 1.14<sup>\*</sup>.* Take expectations on both sides of Lemma 1.12. Using Lemma 1.13 part (2) for the left-hand side, the unbiasedness  $\mathbb{E}[\ell_{t,i}^h] = \ell_{t,i}$  for the comparator term (choosing  $u$  as each coordinate vector), and the bound (4) for the variance term, we obtain

$$\mathbb{E}[L_A(S)] - \min_i L_i(S) \leq \frac{\ln n}{\eta} + \frac{\eta}{2} mn.$$

Substituting  $\eta = \sqrt{2 \ln n / (mn)}$  yields  $\mathbb{E}[L_A(S)] - \min_i L_i(S) \leq \sqrt{2mn \ln n}$ .  $\square$

**Remark 1.15** (Comparison of Regret Rates<sup>\*</sup>). In the *full information* setting, HEDGE achieves regret  $\sqrt{2m \ln n}$ . Moving to partial feedback (EXP3), the regret becomes  $\sqrt{2mn \ln n}$ —an additional  $\sqrt{n}$  factor, which is the price for observing only one arm per round.

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